

REQUEST FOR ACADEMIC DATA COLLABORATION

Predictive Analytics for Emergency Department Demand Forecasting
National University of Computer and Emerging Sciences (FAST-NUCES), Lahore

Institution	FAST-NUCES Lahore — School of Management, Business Analytics
Researchers	Hamza Nadeem (22L-6462) Dania Baig (22L-6484)
Supervisor	Miss Arooj Alvi
Nature of Study	Final Year Project — Non-Commercial Academic Research
Date	March 2026

1. What We Are Doing and Why We Need Your Help

We are final year Business Analytics students at FAST-NUCES Lahore working on a project that uses data to predict medicine and equipment shortages in hospital Emergency Departments before they happen.

The core idea is straightforward: if we know how many patients came to the ED today, what their diagnoses were, and how severe their conditions were — we can forecast how much medicine and equipment the ED will need over the next 7 to 30 days. This helps hospital management stock proactively instead of running short or over-ordering.

To make this research meaningful and practically implementable — not just a theoretical university exercise — we need real patient records from an actual Emergency Department. Your hospital’s data, even in its existing format, is sufficient for our needs.

2. What Data We Need

We need two things, both of which already exist in your hospital’s system in their current form. We are not asking you to prepare, calculate, or reformat anything.

Source 1 — ED Patient Register (Existing Forms / MIS Records)

Your existing patient register — whether maintained in a software system or as physical forms — already contains everything we need. We only require the following fields per patient visit:

Field	Example from Your Register	What We Use It For
Date of Arrival	23/02/2026	To count daily patient volume
Time of Arrival	10:15 AM	To identify peak hours and shift patterns

Field	Example from Your Register	What We Use It For
Date & Time of Discharge	23/02/2026 — 12:30 PM	To calculate how long each patient stayed
Age	22	Patient profile grouping
Gender	Male / Female	Basic demographic split
Diagnosis	Fever, Sore Throat, Respiratory	To link illness type with medicine usage
Medicines Given in ED	Paracetamol, Ondansetron, Drip	Core variable — tells us what was consumed
Disposal / Outcome	Discharged / Admitted / LAMA / Referred	To classify visit severity and resource use

Patient name, CNIC, mobile number, and address are not required. These fields can be removed before sharing for maintaining data anonymity. We only need the clinical and timing information listed above.

Source 2 — Pharmacy Daily Stock / Dispensing Register

Most hospital pharmacies maintain a record of medicines dispensed and stock levels in inventory software. We need the following from that record:

Field	Example	What We Use It For
Date	23/02/2026	Links daily pharmacy usage to patient volume on the same day
Medicine / Item Name	Paracetamol 500mg, Normal Saline 500ml	Item we are forecasting demand for
Units Dispensed / Sold	47 tablets, 12 bottles	Core target variable our models will predict
Opening Stock	200 units	To calculate stockout risk
Closing Stock	153 units	To verify consumption and detect gaps

If your pharmacy system does not track opening and closing stock daily, units dispensed per day is sufficient on its own. We will work with whatever format the register is currently maintained in.

3. How Much Data and What Time Period

Minimum Requirement

- 12 months of ED patient records and pharmacy dispensing data covering the same time period
- Both datasets must cover the same calendar months — this is the only strict condition
- Data can be from any year (2022, 2023, 2024 — whatever is available)
- 24 months of data is preferred but 12 months is fully workable

We understand hospital records may have gaps, missing entries, or inconsistent formats. This is completely acceptable data cleaning and handling missing values is part of our research methodology. We do not expect perfect data.

4. Complete Data Privacy Commitment

We understand patient confidentiality is a legal and ethical obligation. Our commitments are unconditional:

What Will NOT Be Used	Our Commitment
Patient names, CNICs, phone numbers	Hospital name will not appear anywhere in our research if you want anonymity
Any information that identifies a patient	Data used strictly for this academic project only
Financial or billing records	Data will not be shared with any third party
Staff names or performance details	All data deleted upon project completion
Anything beyond what is listed in Section 2	We will sign an NDA if your hospital requires one

5. What Your Hospital Receives in Return

Upon completion of the project, we will provide your hospital with the following:

Deliverables to Your Hospital

- A complete analytical report summarising patient flow patterns, peak demand periods, and medicine usage trends from your own ED data
- An interactive Power BI dashboard showing daily demand forecasts, stock risk alerts, and reorder recommendations for your most used ED medicines and equipment
- A walkthrough session with your management or pharmacy team to explain the findings in plain language
- All outputs are informational only — there is no obligation to adopt or implement anything

6. How to Share the Data

We are flexible on format and will adapt to whatever is available:

- Excel or CSV export from your hospital MIS or pharmacy software
- A scanned or photographed copy of physical registers (patient or pharmacy) — we will digitize it ourselves
- Any database extract your IT team can produce without special effort

We only need one designated point of contact from your hospital. No IT work, reformatting, or preparation is required on your end. One raw export or register copy is all we need.

7. Next Steps

If you are willing to collaborate, the process is simple:

Step	Action	Timeline
1	Confirm willingness to collaborate	When convenient
2	Sign NDA if required by your hospital	Within one week of confirmation
3	Share one copy of ED register + pharmacy dispensing data	One-time only
4	Research team handles all analysis independently	8–10 weeks
5	Final dashboard and report delivered to your hospital	On completion

We recognize that we are the ones who need your cooperation, and we are committed to making this as easy and low-effort as possible for your team. We will work entirely around your availability and comfort level.

Research Team

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Business Analytics — Final Year Project

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